report

Link to the source code: GitHub Repository

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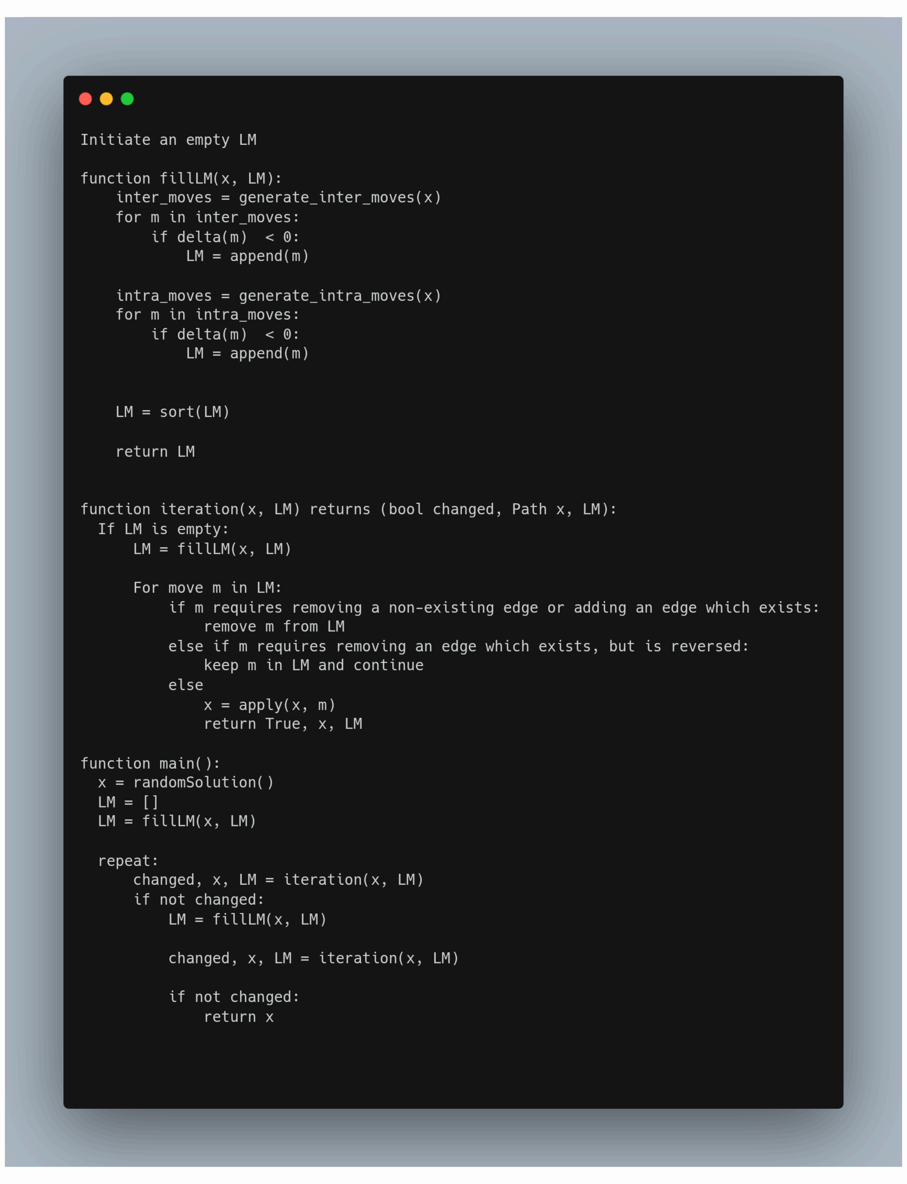
Problem Description

The problem considered in this project involves a set of nodes, each represented by three integer values: two coordinates (x,y)(x, y)(x,y) that define the node's position in a two-dimensional plane, and a cost value associated with the node. The objective is to select exactly 50% of all available nodes (if the total number of nodes is odd, the number of selected nodes is rounded up) and construct a Hamiltonian cycle — that is, a closed path visiting each selected node exactly once and returning to the starting node. The goal is to minimize the sum of two components: 1. The total length of the constructed cycle, and 2. The total cost of all selected nodes The distances between nodes are calculated using the Euclidean metric, rounded to the nearest integer.

Implemented Algorithms



```
Initiate an empty LM

function fillLM(x, LM):
    inter_moves = generate_inter_moves(x)
    for m in inter_moves:
        if delta(m)  < 0:
            LM = append(m)

    intra_moves = generate_intra_moves(x)
    for m in intra_moves:
        if delta(m)  < 0:
            LM = append(m)


    LM = sort(LM)

    return LM


function iteration(x, LM) returns (bool changed, Path x, LM):
  If LM is empty:
      LM = fillLM(x, LM)

      For move m in LM:
          if m requires removing a non-existing edge or adding an edge which exists:
              remove m from LM
          else if m requires removing an edge which exists, but is reversed:
              keep m in LM and continue
          else
              x = apply(x, m)
              return True, x, LM

function main():
  x = randomSolution()
  LM = []
  LM = fillLM(x, LM)

  repeat:
      changed, x, LM = iteration(x, LM)
      if not changed:
          LM = fillLM(x, LM)

          changed, x, LM = iteration(x, LM)

          if not changed:
              return x
```

Results

Comparison of previous methods results

Instance A

Method	Best	Worst	Average
random	235308	292123	264677.17
nn_end	89198	123123	104012.785
nn_anywhere	71488	74410	72635.72
greedy cycle	72639	72639	72639.0
2-Regret Insertion	105852	123428	115474.93
Weighted Sum (α=1.00, β=1.00)	71108	73438	72130.85
greedy_intra:edges_start:greedy	70663	72648	71594.055
greedy_intra:edges_start:random	71564	76328	73528.81
greedy_intra:nodes_start:greedy	70687	73027	71739.395
greedy_intra:nodes_start:random	79677	99035	86909.13
steepest_intra:edges_start:greedy	70510	72614	71463.08
steepest_intra:edges_start:random	71246	78153	73699.995
steepest_intra:nodes_start:greedy	70626	73004	71615.93
steepest_intra:nodes_start:random	81322	95342	87939.335
steepest_intra:edges_start:random	71246	78153	73699.995
steepest_intra:edges_start:random with Cand mech.	73076	85487	77866.91

Method	best	worst	average
steepest_lm_intra:edges_start:random	75245	82747	78856.275

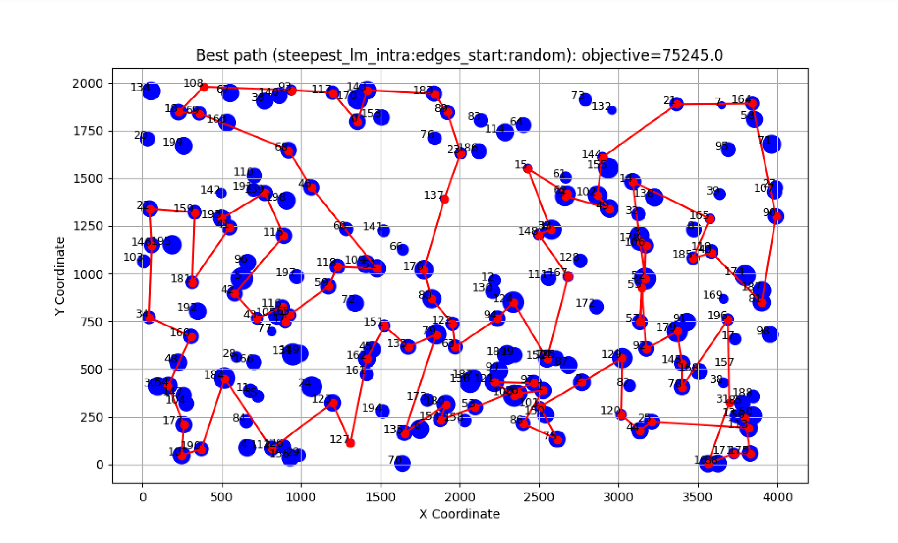
Instance B

Method	Best	Worst	Average
random	193234	234798	214279.235
nn_end	62606	77453	69764.43
nn_anywhere	49001	57324	51400.905
greedy cycle	50243	50243	50243.0
2-Regret Insertion	66505	77072	72454.77
Weighted Sum (α=1.00, β=1.00)	47144	55700	50918.82
greedy_intra:edges_start:greedy	46178	55471	50172.67
greedy_intra:edges_start:random	45537	51687	48154.865
greedy_intra:nodes_start:greedy	46373	55385	50347.525
greedy_intra:nodes_start:random	53417	71474	61453.295
steepest_intra:edges_start:greedy	45867	54814	49907.205
steepest_intra:edges_start:random	45903	51416	48195.845
steepest_intra:nodes_start:greedy	46371	55385	50201.47
steepest_intra:nodes_start:random	55686	71546	62955.885
steepest_intra:edges_start:random	45903	51416	48195.845
steepest_intra:edges_start:random with Cand. Mech.	45798	52670	48474.64

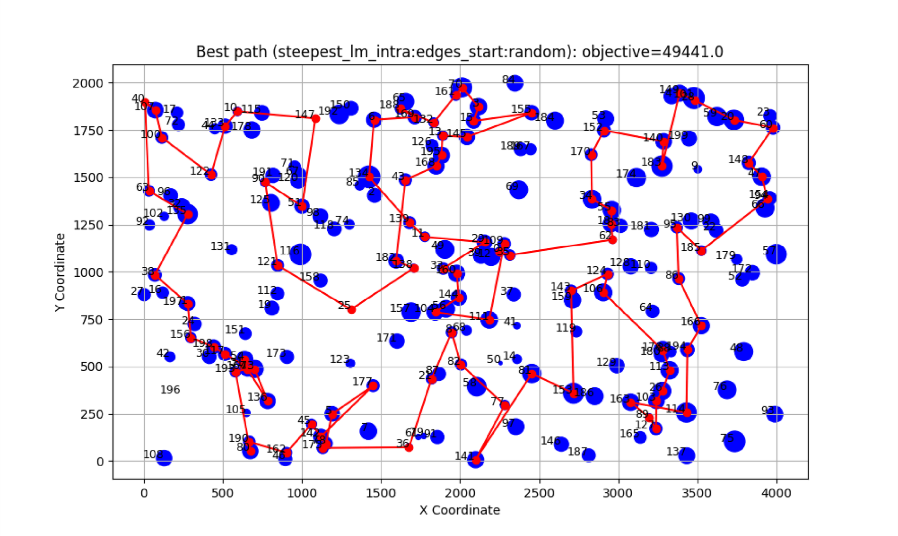
Method	best	worst	average
steepest_lm_intra:edges_start:random	49441	56187	52214.715

Steepest with LM Mechanism results' path

Instance A



Instance B



Time

Instance A

Method	min	max	average
greedy_intra:edges_start:greedy	0.00	0.006	0.003
greedy_intra:edges_start:random	0.062	0.099	0.080
greedy_intra:nodes_start:greedy	0.000	0.005	0.002
greedy_intra:nodes_start:random	0.066	0.118	0.085
steepest_intra:edges_start:greedy	0.001	0.005	0.002
steepest_intra:nodes_start:greedy	0.000	0.005	0.002
steepest_intra:nodes_start:random	0.049	0.142	0.071
steepest_intra:edges_start:random	0.039	0.050	0.045
steepest_intra:edges_start:random with Can. Mech.	0.008	0.019	0.011

Method	min	max	average
steepest_lm_intra:edges_start:random	0.025	0.049	0.039

Instance B

Method	min	max	average
greedy_intra:edges_start:greedy	0.0	0.01	0.004
greedy_intra:edges_start:random	0.064	0.102	0.081
greedy_intra:nodes_start:greedy	0.0	0.007	0.003
greedy_intra:nodes_start:random	0.058	0.107	0.082
steepest_intra:edges_start:greedy	0.001	0.01	0.004
steepest_intra:nodes_start:greedy	0.001	0.007	0.004
steepest_intra:nodes_start:random	0.048	0.086	0.063
steepest_intra:edges_start:random	0.038	0.054	0.045
steepest_intra:edges_start:random with CAn. Mech.	0.009	0.05	0.012

Method	min	max	average
steepest_lm_intra:edges_start:random	0.020	0.053	0.038

Conclusions

The implemented LM mechanism improved time significantly, but the results are slightly worse than the steepest algorithm without LM mechanism. Adding cash mech. to the solution limits our space of the solution giving results slightly worse, but saving time.